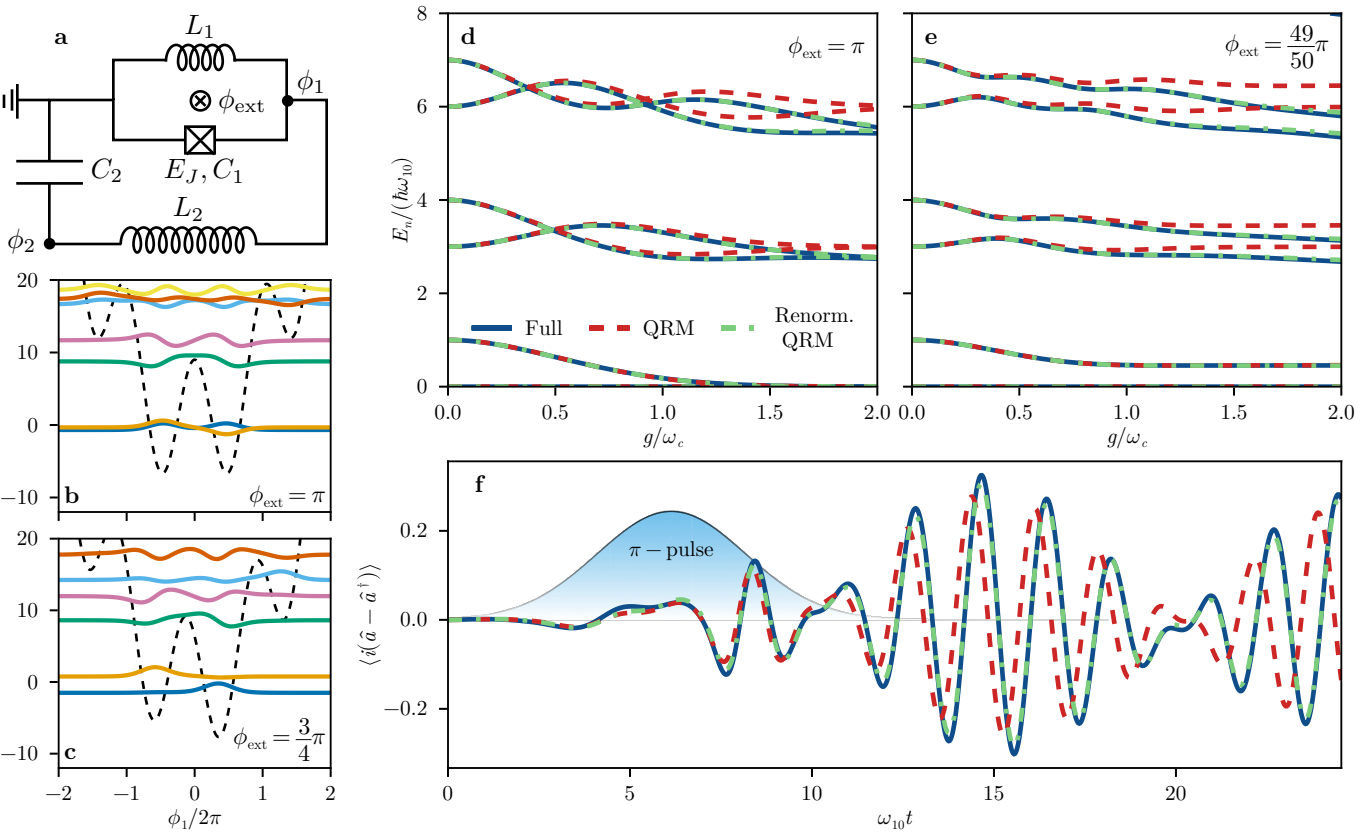
a
L_1
ϕ_1
$\phi_{\rm ext}$
C_2
E_J, C_1
L_2
ϕ_2
b
$\phi_{\rm ext} = \pi$
c
$\phi_{\rm ext} = \frac{3}{4}\pi$
$\phi_1/2\pi$
d
$\phi_{\rm ext} = \pi$
e
$\phi_{\rm ext} = \frac{49}{50}\pi$
$E_n/(\hbar\omega_{10})$
g/ω_c
Full
QRM
Renorm. QRM
f
π − pulse
$\langle \hat{n}(\hat{a} - \hat{a}^\dagger)\rangle$
$\omega_{10}t$